

Paper Validation Report

for DiMag



Correspondence percentages: 77%

Number of experiments found: 2

Experiment 1. Formulation stated: "we perform feature engineering, train and optimize ML models to achieve tenfold cross-validation $R^2 = 0.86$ for prediction of SAR, $R^2 = 0.78$ and $R^2 = 0.75$ for prediction of r_1 and r_2 relaxivities, respectively." **Implementation found:**

`['regression_analysis.py:entire_script()', 'provided_script.py:entire_script()', 'r1_relaxivity_analysis.py:entire_script()', 'analysis_script.py:entire_script()', 'sar_ml_pipeline.py:entire_script()', 'SAR_analysis.py:entire_script()', 'r2_relaxivity_prediction.py:entire_script()']`

Missing components: ['Tenfold cross-validation implementation (not specified in code descriptions)', 'Explicit optimization to achieve R^2 scores of 0.86 for SAR, 0.78 for r_1 , and 0.75 for r_2 '] **Correspondence:** 65.0%

Experiment 2. Formulation stated: "Moreover, we validate the ML models on previously unseen nanoparticles collected independently of the initial database and show accurate predictions of their properties, which makes a case for the use of our models in experimental practice."

Implementation found: `['SAR_analysis.py', 'r1_relaxivity_analysis.py', 'r2_relaxivity_prediction.py']`

Missing components: None **Correspondence:** 90.0%